

SmartNPS360

Complete Functional Test Plan & Feature Inventory

Automatically derived from Flutter native shell source code

Field	Value
Document type	Functional verification test plan (production/release)
Application	SmartNPS360 (WebView shell + native bridges)
App version analyzed	1.0.0+44
Production host	https://smartnps360.com/
Analysis date	2026-07-16
Primary platforms	Android & iOS (release builds)
Execution status baseline	All cases marked Not Tested pending device execution
Architecture note	Flutter hosts InAppWebView; business UI largely web; native owns GPS, auth tokens, push, permissions, media

Objective: Verify every feature, permission-dependent behavior, native integration, user role workflow, and end-to-end path discovered in the codebase works as intended in a production/release environment. Focus is functional correctness, not exploratory bug hunting.

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2. Analysis Methodology & Architecture Summary

This inventory was produced by static analysis of the repository at **lib/**, **android/**, **ios/**, and **pubspec.yaml**. Features were inferred from Dart classes, JS bridge handlers, MethodChannels, AndroidManifest/Info.plist permissions, API URL constants in **AppConfig**, and UI widgets/dialogs. No manual feature list was supplied.

Architecture (as coded)

- SmartNpsApp → WebViewShell on Android/iOS; UnsupportedPlatformScreen elsewhere.
- Web content: <https://smartnps360.com> with UA signature SmartNPS360App.
- Native JS bridge (trusted host only) for media, GPS, push, clock-in, auth, theme, share, download.
- Officer session via Sanctum login/refresh + secure storage; web authEvent for login/logout.
- Duty mode driven by polling POST /api/heartbeat (on_duty / off_duty).
- Background GPS via flutter_background_service (Android FGS) and iOS pinger/SLC.
- Firebase Cloud Messaging + local notifications with custom alert sound.

3. Automatically Detected Feature Inventory

Features below are implemented or enabled in the native shell and/or clearly described by native usage strings (e.g., Daily Activity Reports, visits, profile/ID). Web-only CRUD modules load inside the WebView.

ID	Feature	Intended behavior (from code)	Source
F-01	WebView application shell	Loads https://smartnps360.com in InAppWebView; splash, progress, pull-to-refresh	lib/src/webview/webview_shell.dart
F-02	Officer bottom navigation	Dashboard / TimeSheet / Profile tabs (exact path match)	webview_shell.dart (_BottomItem)
F-03	Officer login (web + Sanctum)	Web login routes + loginWithSanctum native API	auth_session_manager.dart, webview_shell.dart
F-04	Officer registration/sign-up	Detected login-route paths for register/sign-up	auth_session_manager.dart
F-05	Auth event bridge (login/logout)	authEvent JS handler; session persist; logout blocked on duty	webview_shell.dart, AuthRepository
F-06	Token refresh / session continuity	POST /api/auth/refresh; Dio 401 retry interceptor	auth_repository.dart, api_client.dart
F-07	Clock-in / shift attendance gate	prepareClockIn disclosure → permissions → mock check → GPS	clock_in_gate_service.dart, js_bridge.dart
F-08	Foreground GPS for web	getCurrentLocation / location watch with accuracy gate	js_bridge.dart, webview_shell.dart
F-09	On-duty background location tracking	Heartbeat on_duty starts BG GPS + uploads	duty_heartbeat_service.dart, background_location_*
F-10	Off-duty location stop	Heartbeat off_duty stops tracking and drains queue	duty_heartbeat_service.dart
F-11	GPS ping & batch upload	POST /api/gps/ping and /api/gps/batch with Hive queue	background_location_uploader.dart
F-12	Speed-adaptive GPS policy	Adjusts keep/upload frequency by speed band	speed_adaptive_gps_policy.dart
F-13	Batch displacement gate	Skips near-duplicate points for batch upload	batch_displacement_gate.dart
F-14	Mock / fake GPS detection	Blocks clock-in; dialog during duty tracking	mock_location_guard.dart, detection
F-15	Location disclosure consent	Store-safe disclosure before OS permission prompts	location_disclosure_consent.dart, dialogs
F-16	Background location required banner	On-duty UI banner when Always/BG location missing	background_location_required_banner.dart
F-17	Camera capture (takePhoto)	JS bridge → ImagePicker camera for reports/attachments	js_bridge.dart (Info.plist: DAR/visits/ID)
F-18	Gallery image pick	pickImage with photos permission	js_bridge.dart
F-19	File picker	pickFile for attachments	js_bridge.dart
F-20	Share content	share_plus via shareContent handler	js_bridge.dart
F-21	External URL open	url_launcher external application	js_bridge.dart
F-22	File download	downloadFile handler via shell download path	js_bridge.dart, webview_shell.dart
F-23	Device info bridge	getDeviceInfo (brand/model/OS)	js_bridge.dart
F-24	Push notifications (FCM)	Firestore Messaging + local notifications + sound	push_notification_service.dart
F-25	Push token register/delete	POST/DELETE /api/push-token	push_notification_service.dart
F-26	In-app push enable/disable	setPushNotificationsEnabled + prefs + permission-status sync	push_notification_preferences.dart
F-27	Officer announcement push	type=officer_announcement → WebView recipient UUID	officer_announcement_*.dart
F-28	Notification tap deep link	Opens trusted smartnps360 URL (default officer/dashboard)	push_notification_service.dart
F-29	Native permission status sync	POST /api/native-app/permission-status (location, push, battery)	native_permission_status_service.dart
F-30	Battery monitoring upload	Periodic battery % + low power mode while logged in	native_permission_status_service.dart
F-31	Theme sync (light/dark)	themeChanged JS → NativeThemeController	native_theme_controller.dart
F-32	Offline / connectivity handling	OfflineScreen + Retry when no network	offline_screen.dart, webview_shell.dart
F-33	Unsupported platform screen	Non Android/iOS shows mobile-only message	unsupported_platform_screen.dart
F-34	Host allowlist & trusted maps CDN	Blocks untrusted nav; allows Google Maps/Fonts subresources	app_config.dart, webview_shell.dart
F-35	App upgrade reconcile	Clears stale duty/disclosure state after version change	app_upgrade_reconciler.dart

ID	Feature	Intended behavior (from code)	Source
F-36	Open Settings helpers	App settings / location permission settings MethodChannel	permission_settings_helper.dart, MainActivity.kt
F-37	iOS significant location change	Native SLC channel for background duty resilience	ios_significant_location_change_service .dart, AppDelegate.swift
F-38	iOS duty location pinger	Foreground/background position stream for iOS duty	ios_duty_location_pinger.dart
F-39	Logout while on duty blocked	authEvent logout returns logout_blocked_on_duty	webview_shell.dart, duty_heartbeat_service.dart
F-40	Secure credential storage	flutter_secure_storage for tokens/user JSON	auth_repository.dart

4. Automatically Detected Permissions & Dependent Features

ID	Permission	Declared in	Dependent features	When used
P-01	INTERNET / NETWORK_STATE	Android Manifest	WebView + all API calls	Always required
P-02	ACCESS_FINE/COARSE_LOCATION	Android Manifest + runtime	Clock-in GPS, duty tracking, web GPS	Foreground location
P-03	ACCESS_BACKGROUND_LOCATION	Android Q+ runtime	Clock-in gate, on-duty BG tracking	Always allow / all the time
P-04	FOREGROUND_SERVICE_LOCATION	Android Manifest + FGS	BackgroundLocationService notification	On duty
P-05	CAMERA	Android + NSCameraUsageDescription	takePhoto / DAR / visits / ID	When capturing
P-06	RECORD_AUDIO	Android Manifest	WebView media (if used by site)	If web requests mic
P-07	READ_MEDIA_* / storage	Android 13+ / legacy storage	pickImage / attachments	Gallery pick
P-08	POST_NOTIFICATIONS	Android 13+ runtime	FCM / local notifications	Push features
P-09	NSLocationWhenInUse	iOS Info.plist	Foreground GPS	While Using
P-10	NSLocationAlwaysAndWhenInUse	iOS Info.plist	Duty BG tracking / clock-in Always	Always
P-11	NSPhotoLibrary	iOS Info.plist	pickImage / registration images	Photos
P-12	NSUserNotifications	iOS Info.plist + FCM	Push alerts	Notifications
P-13	Battery optimization exemption	Android MethodChannel status	Reported to permission-status API	OEM battery
P-14	Low Power / Battery Saver	Android/iOS power status	Synced to backend	System power mode

Permission-status API fields (synced to backend)

NativePermissionStatusService reports: **foregroundLocation**, **backgroundLocation**, **preciseLocation**, **notifications**, **batteryOptimization**, **push** (in-app toggle), plus **battery_percentage** and **low_power_mode**.

5. Automatically Detected Native Integrations & Packages

ID	Integration / SDK	Functional role
N-01	InAppWebView (flutter_inappwebview)	Primary UI host for smartnps360.com
N-02	Firebase Core + Messaging	Push init, background handler, FCM token
N-03	flutter_local_notifications	Foreground/data push display + custom sound
N-04	Geolocator	GPS stream, permission checks, accuracy
N-05	flutter_background_service	Android FGS location isolate
N-06	permission_handler	Runtime camera/photos/location/notification
N-07	Dio HTTP client	Native APIs with Bearer + refresh retry
N-08	Hive local queue	Offline GPS batch persistence
N-09	flutter_secure_storage	Access/refresh tokens, user, consent keys
N-10	shared_preferences	Push toggle, permission history flags, duty prefs
N-11	image_picker / file_picker	Camera, gallery, files
N-12	share_plus / url_launcher	Share sheet / external browser
N-13	connectivity_plus	Offline detection
N-14	device_info_plus / package_info_plus	Device + app version metadata
N-15	battery_plus	Battery percentage for permission-status
N-16	cupertino_native	iOS native tab bar
N-17	MethodChannel settings	com.smartnps360.app/settings (Android/iOS)
N-18	iOS SLC channels	com.smartnps360.app/ios_slc (+ events)
N-19	Google Maps/Fonts CDN allowlist	Trusted subresource hosts for embedded maps
N-20	Laravel Sanctum APIs	login, refresh, heartbeat, gps, push-token, permission-status

Native API endpoints (AppConfig)

- POST <https://smartnps360.com/api/auth/login>
- POST <https://smartnps360.com/api/auth/refresh>
- GET/POST <https://smartnps360.com/api/heartbeat>
- POST <https://smartnps360.com/api/gps/point|ping|batch>
- POST/DELETE <https://smartnps360.com/api/push-token>
- POST <https://smartnps360.com/api/native-app/permission-status>

6. Automatically Detected User Roles & Workflows

Native code centers on the **Officer** role (`officer_logged_in` flag, `/officer/*` routes, officer announcement pushes). Admin/other web roles are not branched in Flutter; test them only if the production website exposes them through the same WebView.

ID	Role / state	Workflows available	Source
R-01	Unauthenticated visitor	Login/register routes; no duty tracking; push prompts deferred on login	<code>auth_session_manager.isLoginRoute</code>
R-02	Officer (authenticated)	Primary native role: dashboard/timesheet/profile, clock-in, duty GPS, push, permission sync	<code>AuthRepository officer_logged_in; /officer/*</code>
R-03	Officer on duty	Heartbeat on_duty; BG location active; logout blocked until off duty	<code>DutyHeartbeatService</code>
R-04	Officer off duty	Tracking stopped; can logout; GPS upload drained	<code>DutyHeartbeatService.off_duty</code>
R-05	Web admin (if present on site)	Not specially instrumented in native shell; host allowlist only. Verify via web UI if role exists server-side.	No native admin branch

Primary officer workflow (code-derived)

- Open app → WebView loads → Login/Register (auth routes).
- Native Sanctum or `authEvent` persists session tokens.
- Bottom bar: Dashboard ↔ TimeSheet ↔ Profile.
- `prepareClockIn` → disclosure → location permissions → mock check → GPS → web attendance.
- Heartbeat on_duty → background location upload → optional banner if Always missing.
- Heartbeat off_duty → stop tracking → allow logout.
- Push: register token, receive alerts/announcements, tap to navigate.
- Logout: blocked if on duty; otherwise clear session + delete push token + drain GPS.

7. Detected Screens, Routes, JS Bridges & Forms

Native Flutter screens / overlays

- WebViewShell (primary)
- OfflineScreen (Retry)
- UnsupportedPlatformScreen
- Splash overlay (_SplashOverlay)
- Clock-in location disclosure dialog
- Clock-in blocked / Open Settings dialogs (GlassAlertDialog)
- Mock location dialog
- Background location required banner
- Platform bottom bar (Dashboard / TimeSheet / Profile)

Web routes recognized natively

- Auth: /, /officer/login, sign-in, register/sign-up variants
- Logout: /officer/logout or path ending /logout
- Tabs: /officer/dashboard, /officer/timesheet/monthly, /officer/profile
- Generic officer surface: any allowed-host path containing /officer
- Default push destination: /officer/dashboard

JavaScript bridge handlers (trusted host only)

- pickFile, pickImage, takePhoto
- getCurrentLocation, startLocationWatch, clearLocationWatch
- getDeviceInfo, openExternalUrl, shareContent, downloadFile
- getPushNotificationToken, getPushNotificationStatus, setPushNotificationsEnabled
- getBackgroundLocationStatus, prepareClockIn
- themeChanged, loginWithSanctum, authEvent
- iosPopoverFixDebug (iOS diagnostic)

Forms & inputs

Native shell does not define Material text forms for business data. Detected inputs: **loginWithSanctum username/password**; web login/registration/profile/report forms inside WebView; OS permission dialogs; disclosure Accept/Cancel; Open Settings / Cancel on glass dialogs; Offline Retry.

8. Complete Functional Test Cases

Each case uses the required fields: ID, detected feature, source reference, role, preconditions, steps, expected result, status, notes. Status defaults to **Not Tested** for device execution tracking.

Total automatically generated test cases: **77**

Test Case ID	TC-NAV-001
Detected Feature / Integration / Permission	F-01 WebView shell — cold start (release)
Source Code Reference	main.dart; smart_nps_app.dart; webview_shell.dart
User Role	Any (Officer device)
Preconditions	Release/production build installed; network available; device Android or iOS.
Test Steps	1. Launch app from home screen. 2. Observe splash/logo overlay. 3. Wait for WebView to load https://smartnps360.com/.
Expected Result	App initializes Firebase, auth cache, push; splash dismisses; home/login web page loads successfully without debug banners.
Test Status	Not Tested
Notes	Run release: flutter build apk --release / IPA Archive. Debug banner must be absent.

Test Case ID	TC-NAV-002
Detected Feature / Integration / Permission	F-02 Bottom navigation — Dashboard
Source Code Reference	webview_shell.dart _BottomItem.dashboard
User Role	Officer
Preconditions	Officer logged in; URL on an officer surface that shows bottom bar.
Test Steps	1. Tap Dashboard tab. 2. Confirm navigation to /officer/dashboard. 3. Tap Dashboard again (same tab).
Expected Result	WebView loads dashboard; tab highlights; re-tap does not restart unnecessary navigation when already selected.
Test Status	Not Tested
Notes	

Test Case ID	TC-NAV-003
Detected Feature / Integration / Permission	F-02 Bottom navigation — TimeSheet
Source Code Reference	_BottomItem.timesheet → /officer/timesheet/monthly
User Role	Officer
Preconditions	Officer logged in; bottom bar visible.
Test Steps	1. Tap TimeSheet. 2. Verify monthly timesheet page.
Expected Result	Exact path /officer/timesheet/monthly loads; tab selected.
Test Status	Not Tested
Notes	

Test Case ID	TC-NAV-004
Detected Feature / Integration / Permission	F-02 Bottom navigation — Profile
Source Code Reference	_BottomItem.profile → /officer/profile
User Role	Officer
Preconditions	Officer logged in.
Test Steps	1. Tap Profile. 2. Verify profile page content.
Expected Result	Profile URL loads; bottom bar remains correct.
Test Status	Not Tested
Notes	

Test Case ID	TC-NAV-005
Detected Feature / Integration / Permission	F-02 Bottom bar hidden on sub-routes
Source Code Reference	_BottomItem.indexForUri exact match only
User Role	Officer
Preconditions	On dashboard; navigate into a nested officer sub-page (not exact tab URL).
Test Steps	1. Open a nested page from dashboard. 2. Observe bottom bar visibility.
Expected Result	Bottom bar only on exact tab landing URLs — not sibling/subpath pages.
Test Status	Not Tested
Notes	

Test Case ID	TC-NAV-006
Detected Feature / Integration / Permission	F-34 Untrusted host blocked
Source Code Reference	app_config.isAllowedHost; webview_shell navigation
User Role	Officer
Preconditions	App running.
Test Steps	1. Attempt navigation to non-allowed host (via web link if present). 2. Attempt Google Maps tiles/iframe on allowed page.
Expected Result	Untrusted hosts do not replace shell; trusted Google CDN subresources allowed for maps.
Test Status	Not Tested
Notes	

Test Case ID	TC-NAV-007
Detected Feature / Integration / Permission	F-32 Offline screen
Source Code Reference	offline_screen.dart; connectivity_plus
User Role	Any
Preconditions	App open.
Test Steps	1. Enable airplane mode / disable network. 2. Trigger load/navigation. 3. Tap Retry after restoring network.
Expected Result	OfflineScreen shows 'No internet connection' + Retry; Retry reloads when online.
Test Status	Not Tested
Notes	

Test Case ID	TC-NAV-008
Detected Feature / Integration / Permission	F-33 Unsupported platform
Source Code Reference	unsupported_platform_screen.dart
User Role	N/A
Preconditions	Run on web/desktop if available.
Test Steps	1. Launch on non-mobile platform.
Expected Result	Shows 'This app is mobile-only' message.
Test Status	Not Tested
Notes	Optional; primary platforms are Android/iOS.

Test Case ID	TC-AUTH-001
Detected Feature / Integration / Permission	F-03 Officer web login — valid credentials
Source Code Reference	auth_session_manager login routes; authEvent; AuthRepository.saveLogin
User Role	Officer
Preconditions	Valid officer credentials; on login page.
Test Steps	1. Enter valid username/password on web form. 2. Submit login. 3. Observe navigation to officer surface.
Expected Result	Native stores tokens/user; officerLoggedIn=true; dashboard accessible; push/permission sync may start.
Test Status	Not Tested
Notes	

Test Case ID	TC-AUTH-002
Detected Feature / Integration / Permission	F-03 Login — invalid credentials
Source Code Reference	Web form + optional loginWithSanctum
User Role	Unauthenticated
Preconditions	On login page.
Test Steps	1. Enter invalid password. 2. Submit.
Expected Result	Login fails with web/native error; no durable officer session; tokens not treated as logged-in.
Test Status	Not Tested
Notes	

Test Case ID	TC-AUTH-003
Detected Feature / Integration / Permission	F-03 Native Sanctum login bridge
Source Code Reference	loginWithSanctum handler → POST /api/auth/login
User Role	Officer
Preconditions	Trusted host; web triggers flutter_inappwebview.callHandler('loginWithSanctum', {username,password}).
Test Steps	1. Trigger Sanctum login with valid creds. 2. Inspect response {ok:true,hasToken:true}.
Expected Result	Access/refresh tokens saved; untrusted origin denied.
Test Status	Not Tested
Notes	

Test Case ID	TC-AUTH-004
Detected Feature / Integration / Permission	F-04 Registration / sign-up routes
Source Code Reference	isLoginRoute paths officer/register, sign-up, signup
User Role	Unauthenticated
Preconditions	Navigate to registration URL on smartnps360.com.
Test Steps	1. Open register/sign-up. 2. Complete registration form fields required by web. 3. Submit.
Expected Result	Registration workflow completes per web app; treated as auth route (push prompts deferred).
Test Status	Not Tested
Notes	Form fields are web-owned; native treats path as login-family route.

Test Case ID	TC-AUTH-005
Detected Feature / Integration / Permission	F-05 / F-39 Logout when off duty
Source Code Reference	authEvent logout; AuthSessionManager.clearNativeSession
User Role	Officer off duty
Preconditions	Logged in; heartbeat off_duty.
Test Steps	1. Logout from web/profile. 2. Observe native teardown phases.
Expected Result	Tracking stopped; GPS queue drained; FCM token deleted; secure storage cleared; returns to login.
Test Status	Not Tested
Notes	

Test Case ID	TC-AUTH-006
Detected Feature / Integration / Permission	F-39 Logout blocked while on duty
Source Code Reference	webview_shell authEvent logout_blocked_on_duty
User Role	Officer on duty
Preconditions	Heartbeat reports on_duty; tracking active.
Test Steps	1. Attempt logout.
Expected Result	Logout rejected with code logout_blocked_on_duty and message to end shift first; session remains.
Test Status	Not Tested
Notes	

Test Case ID	TC-AUTH-007
Detected Feature / Integration / Permission	F-06 Access token refresh
Source Code Reference	AuthRepository.refreshAccessToken; ApiClient 401 retry
User Role	Officer
Preconditions	Logged in with refresh token; wait for access expiry or force 401 native call.
Test Steps	1. Trigger native API after access token expiry (heartbeat/GPS/push). 2. Observe silent refresh.
Expected Result	POST /api/auth/refresh succeeds; request retried with new Bearer; session continues. Failed refresh clears session.
Test Status	Not Tested
Notes	

Test Case ID	TC-AUTH-008
Detected Feature / Integration / Permission	F-40 Session persistence across kill
Source Code Reference	flutter_secure_storage; warmAccessTokenCache
User Role	Officer
Preconditions	Logged in; force-quit app.
Test Steps	1. Relaunch app. 2. Check if officer session restored without re-login (per web cookie + native tokens).
Expected Result	Native tokens warm from secure storage; authenticated APIs work; web session behavior matches product expectation.
Test Status	Not Tested
Notes	

Test Case ID	TC-CLK-001
Detected Feature / Integration / Permission	F-07 / F-15 Clock-in happy path (permissions granted)
Source Code Reference	prepareClockIn; ClockInGateService; getCurrentLocation for_clock_in
User Role	Officer
Preconditions	Release build; location services ON; Always/background location already granted; disclosure accepted; no mock GPS.
Test Steps	1. From web, start clock-in / shift attendance. 2. Native prepareClockIn runs. 3. GPS returns accurate fix $\leq$ required meters.
Expected Result	canClockIn=true; location payload returned; attendance verification succeeds on web.
Test Status	Not Tested
Notes	

Test Case ID	TC-CLK-002
Detected Feature / Integration / Permission	F-07 Clock-in — disclosure cancel
Source Code Reference	ClockInLocationDisclosureDialog; prepareClockIn
User Role	Officer
Preconditions	Disclosure not accepted; background not ready.
Test Steps	1. Start clock-in. 2. Cancel disclosure dialog.
Expected Result	Clock-in cancelled; canClockIn false; reason clock_in_cancelled / disclosure cancelled.
Test Status	Not Tested
Notes	

Test Case ID	TC-CLK-003
Detected Feature / Integration / Permission	F-07 Clock-in — location services disabled
Source Code Reference	Geolocator.isLocationServiceEnabled
User Role	Officer
Preconditions	Disable device Location Services.
Test Steps	1. Start clock-in.
Expected Result	Blocked with location_services_disabled; Open Settings prompt available.
Test Status	Not Tested
Notes	

Test Case ID	TC-CLK-004
Detected Feature / Integration / Permission	F-14 Clock-in blocked by mock location
Source Code Reference	MockLocationGuard.ensureClearForClockIn
User Role	Officer
Preconditions	Enable mock/fake GPS (developer options).
Test Steps	1. Grant Always location. 2. Attempt clock-in.
Expected Result	Mock location dialog; canClockIn false with reason mock_location until mock disabled.
Test Status	Not Tested
Notes	

Test Case ID	TC-CLK-005
Detected Feature / Integration / Permission	F-08 getCurrentLocation accuracy timeout
Source Code Reference	js_bridge.getCurrentLocation
User Role	Officer
Preconditions	Indoors / poor GPS; required_accuracy_meters tight.
Test Steps	1. Request location with high accuracy requirement. 2. Wait for timeout.
Expected Result	Error fresh_location_unavailable; may include bestAccuracySeenMeters.
Test Status	Not Tested
Notes	

Test Case ID	TC-CLK-006
Detected Feature / Integration / Permission	F-08 Location watch start/clear
Source Code Reference	startLocationWatch / clearLocationWatch handlers
User Role	Officer
Preconditions	Foreground location granted; disclosure accepted.
Test Steps	1. Web starts location watch. 2. Move device; observe updates. 3. Clear watch.
Expected Result	Continuous updates while active; clear stops stream.
Test Status	Not Tested
Notes	

Test Case ID	TC-DUTY-001
Detected Feature / Integration / Permission	F-09 On-duty tracking start
Source Code Reference	DutyHeartbeatService; POST /api/heartbeat; BackgroundLocationController
User Role	Officer on duty
Preconditions	Logged in; Always location; disclosure accepted.
Test Steps	1. Clock in / set on duty on server so heartbeat returns on_duty. 2. Background the app. 3. Verify location notification (Android FGS) / iOS indicator.
Expected Result	Tracking starts; points queued/uploaded; permission-status reflects granted BG location.
Test Status	Not Tested
Notes	

Test Case ID	TC-DUTY-002
Detected Feature / Integration / Permission	F-10 Off-duty stop
Source Code Reference	DutyHeartbeatService off_duty stop
User Role	Officer
Preconditions	Currently on duty with tracking.
Test Steps	1. Clock out / end shift so heartbeat returns off_duty. 2. Observe FGS/notification stop.
Expected Result	Tracking stops; pending batches flush/drain; banner clears.
Test Status	Not Tested
Notes	

Test Case ID	TC-DUTY-003
Detected Feature / Integration / Permission	F-11 GPS ping & batch APIs
Source Code Reference	background_location_uploader.dart; AppConfig gps paths
User Role	Officer on duty
Preconditions	On duty outdoors with movement.
Test Steps	1. Remain on duty for several minutes with movement. 2. Optionally go offline then online.
Expected Result	POST /api/gps/ping and/or /api/gps/batch succeed (200); Hive queue drains after reconnect.
Test Status	Not Tested
Notes	

Test Case ID	TC-DUTY-004
Detected Feature / Integration / Permission	F-12 / F-13 Adaptive keep + displacement
Source Code Reference	speed_adaptive_gps_policy.dart; batch_displacement_gate.dart
User Role	Officer on duty
Preconditions	On duty; walk then ride/drive if safe.
Test Steps	1. Stationary period. 2. Move slowly then faster.
Expected Result	Fewer uploads when stationary/near-duplicate; denser keep when moving per policy (backend receives sensible density).
Test Status	Not Tested
Notes	

Test Case ID	TC-DUTY-005
Detected Feature / Integration / Permission	F-16 Background location banner
Source Code Reference	background_location_required_banner.dart
User Role	Officer on duty
Preconditions	On duty but Only While Using / no Always.
Test Steps	1. Enter on_duty without Always permission. 2. Tap Enable Location on banner.
Expected Result	Banner visible; tapping guides disclosure/settings flow; tracking does not falsely claim ready.
Test Status	Not Tested
Notes	

Test Case ID	TC-DUTY-006
Detected Feature / Integration / Permission	F-37 / F-38 iOS SLC + duty pinger
Source Code Reference	ios_significant_location_change_service.dart; ios_duty_location_pinger.dart; AppDelegate.swift
User Role	Officer on duty (iOS)
Preconditions	iOS Always location; release build; on duty.
Test Steps	1. Background app for extended period. 2. Travel enough for SLC if applicable. 3. Return to foreground.
Expected Result	Location continues best-effort; pinger/SLC channels operate without crash; uploads resume.
Test Status	Not Tested
Notes	iOS BG is best-effort per code comments — verify production behavior on real device.

Test Case ID	TC-MED-001
Detected Feature / Integration / Permission	F-17 takePhoto — permission granted
Source Code Reference	js_bridge.takePhoto; Permission.camera
User Role	Officer
Preconditions	Camera permission allowed.
Test Steps	1. From web feature needing photo (DAR/visit/ID). 2. Capture photo. 3. Confirm. 4. Cancel once.
Expected Result	ok with photo path/name; cancel returns canceled:true.
Test Status	Not Tested
Notes	

Test Case ID	TC-MED-002
Detected Feature / Integration / Permission	F-18 pickImage — gallery
Source Code Reference	js_bridge.pickImage
User Role	Officer
Preconditions	Photos/media permission as required by OS.
Test Steps	1. Pick image from gallery. 2. Cancel picker.
Expected Result	Selected image returned; cancel handled; iOS denial returns permission_denied.
Test Status	Not Tested
Notes	

Test Case ID	TC-MED-003
Detected Feature / Integration / Permission	F-19 pickFile
Source Code Reference	js_bridge.pickFile
User Role	Officer
Preconditions	Trusted host.
Test Steps	1. Trigger file pick. 2. Select a file / cancel.
Expected Result	File metadata returned or canceled:true.
Test Status	Not Tested
Notes	

Test Case ID	TC-MED-004
Detected Feature / Integration / Permission	F-20 shareContent
Source Code Reference	js_bridge.shareContent
User Role	Officer
Preconditions	Trusted host.
Test Steps	1. Trigger share with text. 2. Complete or dismiss share sheet.
Expected Result	Share sheet opens; ok:true on success.
Test Status	Not Tested
Notes	

Test Case ID	TC-MED-005
Detected Feature / Integration / Permission	F-21 openExternalUrl
Source Code Reference	js_bridge.openExternalUrl
User Role	Officer
Preconditions	Trusted host.
Test Steps	1. Open an https URL externally.
Expected Result	launched:true opens in external browser/app.
Test Status	Not Tested
Notes	

Test Case ID	TC-MED-006
Detected Feature / Integration / Permission	F-22 downloadFile
Source Code Reference	downloadFile handler
User Role	Officer
Preconditions	Authenticated; file URL on allowed host if required.
Test Steps	1. Download a report/attachment from web via bridge.
Expected Result	Download completes with ok result; file accessible per platform behavior.
Test Status	Not Tested
Notes	

Test Case ID	TC-MED-007
Detected Feature / Integration / Permission	F-23 getDeviceInfo
Source Code Reference	js_bridge.getDeviceInfo
User Role	Officer
Preconditions	Trusted host.
Test Steps	1. Call getDeviceInfo from web.
Expected Result	Returns platform + device model/SDK or iOS system fields.
Test Status	Not Tested
Notes	

Test Case ID	TC-PUSH-001
Detected Feature / Integration / Permission	F-24 / F-25 Push receive foreground
Source Code Reference	push_notification_service.dart; MainActivity channel smartnps360_default
User Role	Officer
Preconditions	Logged in; notification permission granted; push enabled; token uploaded.
Test Steps	1. Send test FCM from backend while app foregrounded. 2. Observe local notification + sound (alert_sound).
Expected Result	Notification displayed with title/body; custom sound where channel allows.
Test Status	Not Tested
Notes	

Test Case ID	TC-PUSH-002
Detected Feature / Integration / Permission	F-24 Push receive background/killed
Source Code Reference	firebaseMessagingBackgroundHandler
User Role	Officer
Preconditions	App backgrounded or killed; permission granted.
Test Steps	1. Send FCM. 2. Tap notification.
Expected Result	OS/local notification appears; tap opens trusted URL (default /officer/dashboard).
Test Status	Not Tested
Notes	

Test Case ID	TC-PUSH-003
Detected Feature / Integration / Permission	F-26 Toggle push off/on
Source Code Reference	setPushNotificationsEnabled; PushNotificationPreferences
User Role	Officer
Preconditions	Logged in.
Test Steps	1. Disable push via web bridge. 2. Send FCM. 3. Re-enable.
Expected Result	Preference persists; permission-status push field updates; enabled state restores token registration behavior.
Test Status	Not Tested
Notes	

Test Case ID	TC-PUSH-004
Detected Feature / Integration / Permission	F-27 Officer announcement push
Source Code Reference	officer_announcement_push.dart; coordinator
User Role	Officer
Preconditions	Valid recipient_public_id UUID for logged-in officer.
Test Steps	1. Send type=officer_announcement with recipient_public_id. 2. Optional trusted url.
Expected Result	Coordinator forwards UUID to WebView; malformed UUID ignored; untrusted url rejected.
Test Status	Not Tested
Notes	Unit tests exist in test/officer_announcement_*.dart

Test Case ID	TC-PUSH-005
Detected Feature / Integration / Permission	F-28 Notification deep link allowlist
Source Code Reference	parseTrustedDestinationUrl / tap handler
User Role	Officer
Preconditions	Push with url payload.
Test Steps	1. Send push with https://smartnps360.com/... url. 2. Send push with evil.com url.
Expected Result	Trusted URL opens in WebView; untrusted falls back to defaultPushUrl.
Test Status	Not Tested
Notes	

Test Case ID	TC-PERMAPI-001
Detected Feature / Integration / Permission	F-29 Permission-status sync after login
Source Code Reference	NativePermissionStatusService → POST /api/native-app/permission-status
User Role	Officer
Preconditions	Native auth session present.
Test Steps	1. Login. 2. Grant/deny permissions. 3. Resume app from Settings.
Expected Result	Payload includes platform, deviceId, appVersion, build, battery, permissions map, checkedAt; backend receives updates.
Test Status	Not Tested
Notes	

Test Case ID	TC-PERMAPI-002
Detected Feature / Integration / Permission	F-30 Battery monitoring
Source Code Reference	startBatteryMonitoring every 5 minutes
User Role	Officer
Preconditions	Logged in; app active.
Test Steps	1. Keep app open >5 minutes. 2. Change battery level (or wait).
Expected Result	Battery percentage uploaded when changed; stops after logout.
Test Status	Not Tested
Notes	

Test Case ID	TC-THEME-001
Detected Feature / Integration / Permission	F-31 Theme sync
Source Code Reference	themeChanged handler; NativeThemeController
User Role	Officer
Preconditions	Web supports theme toggle.
Test Steps	1. Toggle dark/light in web. 2. Observe native dialogs/scaffold theme.
Expected Result	Native ThemeMode updates; ok:true with isDark.
Test Status	Not Tested
Notes	

Test Case ID	TC-UPG-001
Detected Feature / Integration / Permission	F-35 App upgrade reconcile
Source Code Reference	app_upgrade_reconciler.dart
User Role	Officer
Preconditions	Install prior build with duty flags; upgrade to +44.
Test Steps	1. Upgrade app. 2. Launch.
Expected Result	Stale duty/disclosure/geo unlock cleared safely; no crash before MainActivity channel ready.
Test Status	Not Tested
Notes	

Test Case ID	TC-PERM-P-02-G
Detected Feature / Integration / Permission	P-02 Foreground location — GRANTED
Source Code Reference	Permission.location / Geolocator
User Role	Officer
Preconditions	Foreground location granted in OS Settings; location services on if location.
Test Steps	1. Exercise dependent feature: Clock-in / web GPS / duty. 2. Confirm success path.
Expected Result	Feature completes successfully; permission-status reports granted (or lasting While Using rules for iOS FG).
Test Status	Not Tested
Notes	

Test Case ID	TC-PERM-P-02-D
Detected Feature / Integration / Permission	P-02 Foreground location — DENIED (OS prompt Don't Allow)
Source Code Reference	Permission.location / Geolocator
User Role	Officer
Preconditions	Fresh or reset permission; trigger request.
Test Steps	1. Trigger feature. 2. Deny OS permission prompt.
Expected Result	Feature returns permission_denied (or equivalent); Open Settings education may appear; API may report denied/unknown per history rules.
Test Status	Not Tested
Notes	

Test Case ID	TC-PERM-P-02-S
Detected Feature / Integration / Permission	P-02 Foreground location — DISABLED from Settings
Source Code Reference	PermissionSettingsHelper + Settings
User Role	Officer
Preconditions	Permission previously granted.
Test Steps	1. Open Settings and disable permission. 2. Return to app. 3. Retry feature.
Expected Result	Feature blocked; settings prompt / banner as designed; permission-status sync updates on resume.
Test Status	Not Tested
Notes	

Test Case ID	TC-PERM-P-02-R
Detected Feature / Integration / Permission	P-02 Foreground location — RE-GRANTED after deny/disable
Source Code Reference	recheckAfterAppResume; syncLocationPermissionsOnResume
User Role	Officer
Preconditions	Permission currently denied/disabled.
Test Steps	1. Open Settings from in-app prompt. 2. Grant permission. 3. Return to app. 4. Retry feature.
Expected Result	App detects new grant; clock-in/duty/push proceeds; failure dialog not shown if now ready.
Test Status	Not Tested
Notes	

Test Case ID	TC-PERM-P-03-G
Detected Feature / Integration / Permission	P-03 Background / Always location — GRANTED
Source Code Reference	ACCESS_BACKGROUND_LOCATION / Always
User Role	Officer
Preconditions	Background / Always location granted in OS Settings; location services on if location.
Test Steps	1. Exercise dependent feature: Clock-in gate + on-duty BG tracking. 2. Confirm success path.
Expected Result	Feature completes successfully; permission-status reports granted (or lasting While Using rules for iOS FG).
Test Status	Not Tested
Notes	

Test Case ID	TC-PERM-P-03-D
Detected Feature / Integration / Permission	P-03 Background / Always location — DENIED (OS prompt Don't Allow)
Source Code Reference	ACCESS_BACKGROUND_LOCATION / Always
User Role	Officer
Preconditions	Fresh or reset permission; trigger request.
Test Steps	1. Trigger feature. 2. Deny OS permission prompt.
Expected Result	Feature returns permission_denied (or equivalent); Open Settings education may appear; API may report denied/unknown per history rules.
Test Status	Not Tested
Notes	

Test Case ID	TC-PERM-P-03-S
Detected Feature / Integration / Permission	P-03 Background / Always location — DISABLED from Settings
Source Code Reference	PermissionSettingsHelper + Settings
User Role	Officer
Preconditions	Permission previously granted.
Test Steps	1. Open Settings and disable permission. 2. Return to app. 3. Retry feature.
Expected Result	Feature blocked; settings prompt / banner as designed; permission-status sync updates on resume.
Test Status	Not Tested
Notes	

Test Case ID	TC-PERM-P-03-R
Detected Feature / Integration / Permission	P-03 Background / Always location — RE-GRANTED after deny/disable
Source Code Reference	recheckAfterAppResume; syncLocationPermissionsOnResume
User Role	Officer
Preconditions	Permission currently denied/disabled.
Test Steps	1. Open Settings from in-app prompt. 2. Grant permission. 3. Return to app. 4. Retry feature.
Expected Result	App detects new grant; clock-in/duty/push proceeds; failure dialog not shown if now ready.
Test Status	Not Tested
Notes	

Test Case ID	TC-PERM-P-05-G
Detected Feature / Integration / Permission	P-05 Camera — GRANTED
Source Code Reference	Permission.camera
User Role	Officer
Preconditions	Camera granted in OS Settings; location services on if location.
Test Steps	1. Exercise dependent feature: takePhoto. 2. Confirm success path.
Expected Result	Feature completes successfully; permission-status reports granted (or lasting While Using rules for iOS FG).
Test Status	Not Tested
Notes	

Test Case ID	TC-PERM-P-05-D
Detected Feature / Integration / Permission	P-05 Camera — DENIED (OS prompt Don't Allow)
Source Code Reference	Permission.camera
User Role	Officer
Preconditions	Fresh or reset permission; trigger request.
Test Steps	1. Trigger feature. 2. Deny OS permission prompt.
Expected Result	Feature returns permission_denied (or equivalent); Open Settings education may appear; API may report denied/unknown per history rules.
Test Status	Not Tested
Notes	

Test Case ID	TC-PERM-P-05-S
Detected Feature / Integration / Permission	P-05 Camera — DISABLED from Settings
Source Code Reference	PermissionSettingsHelper + Settings
User Role	Officer
Preconditions	Permission previously granted.
Test Steps	1. Open Settings and disable permission. 2. Return to app. 3. Retry feature.
Expected Result	Feature blocked; settings prompt / banner as designed; permission-status sync updates on resume.
Test Status	Not Tested
Notes	

Test Case ID	TC-PERM-P-05-R
Detected Feature / Integration / Permission	P-05 Camera — RE-GRANTED after deny/disable
Source Code Reference	recheckAfterAppResume; syncLocationPermissionsOnResume
User Role	Officer
Preconditions	Permission currently denied/disabled.
Test Steps	1. Open Settings from in-app prompt. 2. Grant permission. 3. Return to app. 4. Retry feature.
Expected Result	App detects new grant; clock-in/duty/push proceeds; failure dialog not shown if now ready.
Test Status	Not Tested
Notes	

Test Case ID	TC-PERM-P-07-G
Detected Feature / Integration / Permission	P-07 Photos / media — GRANTED
Source Code Reference	Permission.photos / READ_MEDIA_*
User Role	Officer
Preconditions	Photos / media granted in OS Settings; location services on if location.
Test Steps	1. Exercise dependent feature: pickImage. 2. Confirm success path.
Expected Result	Feature completes successfully; permission-status reports granted (or lasting While Using rules for iOS FG).
Test Status	Not Tested
Notes	

Test Case ID	TC-PERM-P-07-D
Detected Feature / Integration / Permission	P-07 Photos / media — DENIED (OS prompt Don't Allow)
Source Code Reference	Permission.photos / READ_MEDIA_*
User Role	Officer
Preconditions	Fresh or reset permission; trigger request.
Test Steps	1. Trigger feature. 2. Deny OS permission prompt.
Expected Result	Feature returns permission_denied (or equivalent); Open Settings education may appear; API may report denied/unknown per history rules.
Test Status	Not Tested
Notes	

Test Case ID	TC-PERM-P-07-S
Detected Feature / Integration / Permission	P-07 Photos / media — DISABLED from Settings
Source Code Reference	PermissionSettingsHelper + Settings
User Role	Officer
Preconditions	Permission previously granted.
Test Steps	1. Open Settings and disable permission. 2. Return to app. 3. Retry feature.
Expected Result	Feature blocked; settings prompt / banner as designed; permission-status sync updates on resume.
Test Status	Not Tested
Notes	

Test Case ID	TC-PERM-P-07-R
Detected Feature / Integration / Permission	P-07 Photos / media — RE-GRANTED after deny/disable
Source Code Reference	recheckAfterAppResume; syncLocationPermissionsOnResume
User Role	Officer
Preconditions	Permission currently denied/disabled.
Test Steps	1. Open Settings from in-app prompt. 2. Grant permission. 3. Return to app. 4. Retry feature.
Expected Result	App detects new grant; clock-in/duty/push proceeds; failure dialog not shown if now ready.
Test Status	Not Tested
Notes	

Test Case ID	TC-PERM-P-08-G
Detected Feature / Integration / Permission	P-08 Notifications — GRANTED
Source Code Reference	POST_NOTIFICATIONS / FCM auth
User Role	Officer
Preconditions	Notifications granted in OS Settings; location services on if location.
Test Steps	1. Exercise dependent feature: FCM display + prompt. 2. Confirm success path.
Expected Result	Feature completes successfully; permission-status reports granted (or lasting While Using rules for iOS FG).
Test Status	Not Tested
Notes	

Test Case ID	TC-PERM-P-08-D
Detected Feature / Integration / Permission	P-08 Notifications — DENIED (OS prompt Don't Allow)
Source Code Reference	POST_NOTIFICATIONS / FCM auth
User Role	Officer
Preconditions	Fresh or reset permission; trigger request.
Test Steps	1. Trigger feature. 2. Deny OS permission prompt.
Expected Result	Feature returns permission_denied (or equivalent); Open Settings education may appear; API may report denied/unknown per history rules.
Test Status	Not Tested
Notes	

Test Case ID	TC-PERM-P-08-S
Detected Feature / Integration / Permission	P-08 Notifications — DISABLED from Settings
Source Code Reference	PermissionSettingsHelper + Settings
User Role	Officer
Preconditions	Permission previously granted.
Test Steps	1. Open Settings and disable permission. 2. Return to app. 3. Retry feature.
Expected Result	Feature blocked; settings prompt / banner as designed; permission-status sync updates on resume.
Test Status	Not Tested
Notes	

Test Case ID	TC-PERM-P-08-R
Detected Feature / Integration / Permission	P-08 Notifications — RE-GRANTED after deny/disable
Source Code Reference	recheckAfterAppResume; syncLocationPermissionsOnResume
User Role	Officer
Preconditions	Permission currently denied/disabled.
Test Steps	1. Open Settings from in-app prompt. 2. Grant permission. 3. Return to app. 4. Retry feature.
Expected Result	App detects new grant; clock-in/duty/push proceeds; failure dialog not shown if now ready.
Test Status	Not Tested
Notes	

Test Case ID	TC-PERM-P02-ONETIME
Detected Feature / Integration / Permission	P-02 Android 'Allow only this time' treated insufficient
Source Code Reference	MainActivity.hasOneTimeLocationPermission; NativePermissionStatusService
User Role	Officer (Android)
Preconditions	Android 11+.
Test Steps	1. Choose Allow only this time for location. 2. Attempt clock-in / check permission-status.
Expected Result	Clock-in requires lasting background readiness; API may report foregroundLocation denied for one-time; user guided to lasting Always.
Test Status	Not Tested
Notes	

Test Case ID	TC-NAT-001
Detected Feature / Integration / Permission	N-17 MethodChannel openAppSettings / location settings
Source Code Reference	MainActivity.kt; permission_settings_helper.dart
User Role	Officer
Preconditions	Release build on Android.
Test Steps	1. Trigger Open Settings from clock-in dialog. 2. Confirm correct Settings screen. 3. Return via back.
Expected Result	Settings opens without crash; resume recheck runs.
Test Status	Not Tested
Notes	

Test Case ID	TC-NAT-002
Detected Feature / Integration / Permission	N-02/N-03 Firebase + local notifications release
Source Code Reference	main.dart Firebase.initializeApp; google-services
User Role	Officer
Preconditions	Production Firebase project config in release build.
Test Steps	1. Confirm FCM token obtained. 2. Receive production push.
Expected Result	Token uploaded; notification channel smartnps360_default exists; sound works.
Test Status	Not Tested
Notes	

Test Case ID	TC-NAT-003
Detected Feature / Integration / Permission	N-05 Android foreground service location
Source Code Reference	AndroidManifest BackgroundService; BackgroundLocationService
User Role	Officer on duty
Preconditions	On duty; Always location.
Test Steps	1. Background app. 2. Check ongoing 'Sharing live location' notification.
Expected Result	FGS notification present (channel smartnps360_location); tracking continues.
Test Status	Not Tested
Notes	

Test Case ID	TC-NAT-004
Detected Feature / Integration / Permission	N-08 Hive GPS queue survives process death
Source Code Reference	background_location_uploader.dart
User Role	Officer on duty
Preconditions	On duty; briefly kill network then force-stop carefully.
Test Steps	1. Generate points offline. 2. Restore network and relaunch.
Expected Result	Queued batches upload when possible; no silent permanent loss without drain/discard rules.
Test Status	Not Tested
Notes	

Test Case ID	TC-NAT-005
Detected Feature / Integration / Permission	N-18 iOS significant location change channel
Source Code Reference	AppDelegate.swift ios_slc
User Role	Officer on duty (iOS)
Preconditions	Always location; release IPA.
Test Steps	1. Enable duty tracking. 2. Background and move.
Expected Result	SLC monitoring registers without Flutter crash; events delivered when OS wakes app.
Test Status	Not Tested
Notes	

Test Case ID	TC-NAT-006
Detected Feature / Integration / Permission	N-14 Low power mode change event
Source Code Reference	MainActivity lowPowerModeChanged; AppDelegate
User Role	Officer
Preconditions	Logged in.
Test Steps	1. Toggle Battery Saver / Low Power Mode. 2. Observe permission-status sync.
Expected Result	low_power_mode field updates to enabled/disabled.
Test Status	Not Tested
Notes	

Test Case ID	TC-E2E-001
Detected Feature / Integration / Permission	E2E Officer day: login → clock-in → duty track → clock-out → logout
Source Code Reference	Multiple modules (auth, clock-in, heartbeat, GPS, logout)
User Role	Officer
Preconditions	Release build; Always location; notifications; production backend.
Test Steps	1. Login. 2. PrepareClockIn + clock in. 3. Work shift with app backgrounded. 4. Clock out. 5. Logout.
Expected Result	Each phase succeeds; GPS visible server-side; logout allowed only after off duty; tokens cleared.
Test Status	Not Tested
Notes	

Test Case ID	TC-E2E-002
Detected Feature / Integration / Permission	E2E Push announcement while on duty
Source Code Reference	push + announcement coordinator + webview
User Role	Officer on duty
Preconditions	On duty; push granted.
Test Steps	1. Send officer_announcement. 2. Tap notification. 3. Confirm WebView receives recipient id / navigates.
Expected Result	Announcement handled without stopping duty tracking.
Test Status	Not Tested
Notes	

Test Case ID	TC-E2E-003
Detected Feature / Integration / Permission	E2E Media attachment on report while authenticated
Source Code Reference	takePhoto/pickImage + web upload
User Role	Officer
Preconditions	Camera/photos granted.
Test Steps	1. Open Daily Activity Report or visit evidence flow on web. 2. Capture/attach photo. 3. Submit form.
Expected Result	Native picker returns path; web upload succeeds; record visible.
Test Status	Not Tested
Notes	Web form fields are server-rendered; native provides capture bridge only.

Test Case ID	TC-E2E-004
Detected Feature / Integration / Permission	E2E Timesheet monthly review
Source Code Reference	Bottom tab timesheet URL
User Role	Officer
Preconditions	Logged in; has historical attendance.
Test Steps	1. Open TimeSheet tab. 2. Review monthly entries. 3. Navigate back to Dashboard.
Expected Result	Timesheet loads; navigation stable; no auth loss.
Test Status	Not Tested
Notes	

Test Case ID	TC-E2E-005
Detected Feature / Integration / Permission	E2E Profile update with image
Source Code Reference	/officer/profile + pickImage
User Role	Officer
Preconditions	Photos permission.
Test Steps	1. Open Profile. 2. Update fields / attach profile or ID image. 3. Save.
Expected Result	Profile updates persist; image pick works.
Test Status	Not Tested
Notes	

Test Case ID	TC-CRUD-001
Detected Feature / Integration / Permission	Web CRUD via shell — create attendance/visit/report records
Source Code Reference	Web app at smartnps360.com (hosted) + native bridges
User Role	Officer
Preconditions	Authenticated.
Test Steps	1. Create a new report/visit/attendance as provided by web UI. 2. Read/list it. 3. Update if UI allows. 4. Delete/cancel if UI allows.
Expected Result	CRUD completes through WebView; native GPS/camera used where required.
Test Status	Not Tested
Notes	Native repo does not implement server CRUD UI; verify against production web modules.

9. Permission-State Testing Matrix

For each permission, verify dependent feature behavior across four states. Detailed steps are in TC-PERM-* cases above; this matrix is the execution tracker.

Perm	Permission	Feature	State	Expected behavior	Status
P-02	Foreground location	GPS / clock-in / duty FG	Granted/Enabled	Feature succeeds; API granted/enabled as applicable	Not Tested
P-02	Foreground location	GPS / clock-in / duty FG	Denied on prompt	Feature errors; settings education may show	Not Tested
P-02	Foreground location	GPS / clock-in / duty FG	Disabled in Settings	Feature blocked; resume sync updates API	Not Tested
P-02	Foreground location	GPS / clock-in / duty FG	Re-granted after deny	Feature recovers without requiring reinstall	Not Tested
P-03	Background/Always location	Clock-in + BG duty tracking	Granted/Enabled	Feature succeeds; API granted/enabled as applicable	Not Tested
P-03	Background/Always location	Clock-in + BG duty tracking	Denied on prompt	Feature errors; settings education may show	Not Tested
P-03	Background/Always location	Clock-in + BG duty tracking	Disabled in Settings	Feature blocked; resume sync updates API	Not Tested
P-03	Background/Always location	Clock-in + BG duty tracking	Re-granted after deny	Feature recovers without requiring reinstall	Not Tested
P-05	Camera	takePhoto / evidence	Granted/Enabled	Feature succeeds; API granted/enabled as applicable	Not Tested
P-05	Camera	takePhoto / evidence	Denied on prompt	Feature errors; settings education may show	Not Tested
P-05	Camera	takePhoto / evidence	Disabled in Settings	Feature blocked; resume sync updates API	Not Tested
P-05	Camera	takePhoto / evidence	Re-granted after deny	Feature recovers without requiring reinstall	Not Tested
P-07	Photos/media	pickImage / attachments	Granted/Enabled	Feature succeeds; API granted/enabled as applicable	Not Tested
P-07	Photos/media	pickImage / attachments	Denied on prompt	Feature errors; settings education may show	Not Tested
P-07	Photos/media	pickImage / attachments	Disabled in Settings	Feature blocked; resume sync updates API	Not Tested
P-07	Photos/media	pickImage / attachments	Re-granted after deny	Feature recovers without requiring reinstall	Not Tested
P-08	Notifications	FCM alerts	Granted/Enabled	Feature succeeds; API granted/enabled as applicable	Not Tested
P-08	Notifications	FCM alerts	Denied on prompt	Feature errors; settings education may show	Not Tested
P-08	Notifications	FCM alerts	Disabled in Settings	Feature blocked; resume sync updates API	Not Tested
P-08	Notifications	FCM alerts	Re-granted after deny	Feature recovers without requiring reinstall	Not Tested
P-13	Battery optimization (Android)	Reported status / tracking resilience	Granted/Enabled	Feature succeeds; API granted/enabled as applicable	Not Tested
P-13	Battery optimization (Android)	Reported status / tracking resilience	Denied on prompt	Feature errors; settings education may show	Not Tested
P-13	Battery optimization (Android)	Reported status / tracking resilience	Disabled in Settings	Feature blocked; resume sync updates API	Not Tested
P-13	Battery optimization (Android)	Reported status / tracking resilience	Re-granted after deny	Feature recovers without requiring reinstall	Not Tested

10. Native Integration Testing Checklist

Done	ID	Release verification item
■	N-01	InAppWebView loads production site; pull-to-refresh works
■	N-02	Firebase initializes in release; FCM token obtained
■	N-03	Local notifications + alert_sound channel
■	N-04	Geolocator accuracy gating for clock-in
■	N-05	Android FGS 'Sharing live location' while on duty
■	N-06	Runtime permission prompts appear only after disclosure
■	N-07	Dio Bearer + 401 refresh retry on native APIs
■	N-08	Hive GPS queue flush after offline
■	N-09	Secure storage survives app restart
■	N-10	SharedPreferences push/permission history
■	N-11	Camera / gallery / file pickers
■	N-12	Share sheet / external URL
■	N-13	OfflineScreen on connectivity loss
■	N-14/15	Device id + battery in permission-status
■	N-16	iOS Cupertino tab bar
■	N-17	Settings MethodChannel opens correct screens
■	N-18	iOS SLC channels under Always location
■	N-19	Embedded Google Maps render in WebView
■	N-20	All AppConfig API endpoints reachable with prod auth

11. End-to-End Workflow Testing Checklist

Done	ID	Workflow
■	E2E-01	Fresh install → login → dashboard
■	E2E-02	Register/sign-up path (if enabled in prod)
■	E2E-03	Clock-in with full permission ladder (disclosure→FG→Always)
■	E2E-04	On-duty background tracking for ≥15 minutes
■	E2E-05	Offline duty → reconnect → batch upload
■	E2E-06	Mock GPS blocked during clock-in
■	E2E-07	Logout blocked while on duty; allowed after off duty
■	E2E-08	Push receive (FG/BG/killed) + tap navigation
■	E2E-09	Officer announcement push with valid UUID
■	E2E-10	Camera evidence attach + submit web form
■	E2E-11	Gallery attach for profile/ID
■	E2E-12	Timesheet monthly review via bottom tab
■	E2E-13	Profile update + theme toggle
■	E2E-14	Token refresh during long duty session
■	E2E-15	App upgrade reconcile from previous build
■	E2E-16	Permission revoke mid-duty → banner → re-grant → recover

12. Production / Release Testing Checklist

Done	Production gate
■	Build release APK/AAB and App Store IPA (not debug)
■	Confirm ProGuard/R8 does not break MethodChannels or Firebase
■	Verify production Firebase options / GoogleService-Info.plist
■	Verify deep links / notification channel IDs match prod
■	Test on low-end Android + recent Pixel/Samsung; iPhone + recent iOS
■	Test OEM battery restrictions (Xiaomi/Vivo/etc.) with Always location
■	Confirm UA contains SmartNPS360App for web detection
■	Confirm allowedHosts only smartnps360.com / www
■	Confirm no debug logging of tokens in release
■	Validate Info.plist / Play Store disclosure strings match actual BG location use
■	Regression after force-stop, reboot, airplane mode
■	Confirm version 1.0.0+44 (or release candidate) reported in permission-status

13. Test Execution Summary

Metric	Value
Features inventoried	40
Permissions inventoried	14
Native integrations inventoried	20
Roles / duty states inventoried	5
Functional test cases generated	77
Passed	0 (pending execution)
Failed	0 (pending execution)
Not Tested	77
Blocked	0
Environment for execution	Production/release Android + iOS devices against smartnps360.com
Automated unit tests present	officer_announcement_push_test.dart, officer_announcement_coordinator_test.dart
Overall readiness	Inventory & plan complete — execute TC-* on release builds and update Status column

Sign-off

Role	Signature / notes
Prepared by (analysis)	Automated code analysis — Cursor Agent
QA executor	_____
Date executed	_____
Android build / device	_____
iOS build / device	_____
Result approval	_____

End of document. Update each test case Status field to Pass/Fail during execution. Re-generate or annotate this PDF as the living QA record for SmartNPS360 native shell releases.